

Senior Project

AI-Powered Quote & Proposal Generation Tool for CRM Platforms

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AI-Powered Quote & Proposal Generation Tool for CRM Platforms

By

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Abstract

Manual quote and proposal generation remains one of the most time-consuming and error-prone bottlenecks in modern sales workflows. Sales representatives routinely spend two to four hours per proposal manually copying data from CRM systems into document templates, leading to pricing errors, outdated compliance clauses, and inconsistent branding. Industry data indicates that 65% of sales representatives report that manual processes slow deal closure, while 23% of proposals generated without automation contain errors. Existing commercial solutions such as Salesforce CPQ, PandaDoc, and HubSpot Quotes address portions of this problem but carry significant licensing costs ranging from \$75 to \$150 per user per month, making them prohibitive for small and mid-sized businesses. Furthermore, no existing tool simultaneously addresses both international data protection requirements (SOC 2, GDPR) and Pakistani procurement regulations (PPRA).

This project presents QuoteForge, an AI-Powered Quote and Proposal Generation Tool designed to automate the end-to-end creation, formatting, and delivery of professional sales proposals directly from CRM data. The system employs a Retrieval-Augmented Generation (RAG) approach combined with a Large Language Model (LLM) to generate context-aware, factually grounded proposal content. A layered architecture comprising seven core modules handles CRM data retrieval via OAuth 2.0, business rule application through a configurable pricing engine, AI-based text generation, branded document rendering in PDF and DOCX formats, and automated delivery with CRM activity logging. An administrative web portal built with React and Tailwind CSS provides configuration interfaces for templates, pricing rules, AI prompt management, and role-based access control.

Evaluation against the defined requirements demonstrates that the system reduces proposal generation time from hours to under ten seconds, eliminates manual data re-entry errors, and enforces regional compliance automatically. The RAG-based generation architecture ensures that AI outputs remain grounded in verified data sources, achieving measurably lower hallucination rates compared to unconstrained generation. QuoteForge offers a cost-effective, scalable alternative to commercial CPQ tools while providing dual-market compliance coverage for both international and Pakistani organizations.

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Revision History

Name	Date	Reason For Changes	Version
Faraz Ali	28 December 2025	Initial SRS draft	1.0
Fahad Saleem	03 January 2026	Formatting and structure alignment	1.1
Agha Zain Nadir	20 January 2026	Use cases, architecture details, compliance sections	2.0
Saad Khalid	05 February 2026	Frontend screenshots, UI specifications, test cases	2.1
Full Team	12 February 2026	Final comprehensive revision per university template	3.0

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Chapter 1. Introduction

1.1 Introduction

In contemporary business-to-business (B2B) sales environments, the ability to generate accurate, professional, and timely proposals is directly correlated with deal closure rates and revenue performance. Customer Relationship Management (CRM) platforms such as Salesforce and HubSpot have become the operational backbone for managing sales pipelines, tracking customer interactions, and storing deal data. However, despite the maturity of CRM ecosystems, the critical last-mile task of converting deal data into formatted, compliant sales documents remains overwhelmingly manual.

Sales representatives typically spend two to four hours per deal creating quotes and proposals by manually extracting data from CRM records, applying pricing calculations, selecting appropriate legal and compliance clauses, and formatting documents according to organizational branding standards. This manual process introduces significant risks: pricing errors from miscalculated discounts or tax rates, inclusion of outdated terms and conditions, inconsistent branding across documents, and delays that allow competitors to present their proposals first. Research indicates that 67% of sales representatives' time is consumed by non-selling activities, with proposal creation being one of the largest contributors.

Commercial Configure-Price-Quote (CPQ) solutions address portions of this problem but present their own challenges. Enterprise platforms such as Salesforce CPQ carry licensing costs of \$75 to \$150 per user per month and require extensive implementation and customization periods. These costs make them inaccessible to small and mid-sized businesses. Furthermore, existing solutions lack integrated AI-based content generation capabilities and fail to address multi-jurisdictional compliance requirements, particularly for organizations operating across both international markets (requiring SOC 2 and GDPR compliance) and the Pakistani market (requiring adherence to Public Procurement Regulatory Authority regulations).

QuoteForge addresses these challenges through an AI-powered automation system that integrates directly with CRM platforms to generate professional, compliant, and brand-consistent quotes and proposals in seconds rather than hours. The system employs a Retrieval-Augmented Generation (RAG) architecture combined with a Large Language Model (LLM) to produce context-aware document content that is grounded in verified data sources. A configurable rules and pricing engine ensures accurate calculations and regional compliance, while a branded document rendering pipeline outputs production-ready PDF and DOCX files. The entire system is managed through an intuitive web-based administrative portal.

1.2 Objectives

The primary objectives of this project are:

1. Develop a CRM-integrated automation system that retrieves deal, account, contact, and product line item data from Salesforce and HubSpot using secure OAuth 2.0 authentication.
2. Implement an AI-based content generation engine using Retrieval-Augmented Generation (RAG) combined with a Large Language Model (LLM) to produce context-aware, factually grounded proposal sections including cover letters, scope descriptions, pricing summaries, deliverables, and terms and conditions.
3. Build a configurable rules and pricing engine that applies discount thresholds, tax calculations, and region-specific compliance clauses (SOC 2, GDPR, PPRA) based on deal context.
4. Create a document rendering pipeline that generates professionally formatted quotes and proposals in PDF and DOCX formats using organization-approved templates and branding assets.
5. Implement automated document delivery through CRM email functionality or SMTP services with full activity logging back to the CRM system.
6. Develop a web-based administrative portal using React and Tailwind CSS for managing templates, pricing rules, AI prompts, CRM integrations, user access, and system settings.
7. Ensure the system operates within acceptable performance constraints, generating complete proposals in under ten seconds for typical deal configurations.
8. Provide dual-market compliance support covering both international standards (SOC 2, GDPR) and Pakistani procurement regulations (PPRA).

1.3 Problem Statement

Manual quote and proposal generation in CRM-driven sales environments causes significant operational inefficiencies, document inconsistencies, and revenue loss. Sales teams spend disproportionate time on administrative document creation rather than revenue-generating selling activities. Copy-paste workflows between CRM systems and document editors introduce pricing errors, outdated compliance clauses, and brand inconsistencies that damage professional credibility and client trust.

Existing commercial CPQ and proposal automation tools partially address these challenges but remain cost-prohibitive for small and mid-sized businesses, require extensive implementation timelines, and lack integrated AI content generation capabilities. Critically, no existing solution simultaneously addresses the compliance

requirements of both international markets (SOC 2, GDPR) and the Pakistani procurement environment (PPRA), creating a significant gap for organizations operating across these jurisdictions.

QuoteForge is an AI-powered, CRM-integrated automation system that eliminates manual proposal creation by combining secure CRM data retrieval, intelligent pricing and compliance rule application, RAG-based AI content generation, branded document rendering, and automated delivery with CRM logging. The system provides a cost-effective, scalable, and dual-compliant alternative to commercial CPQ platforms while achieving proposal generation times measured in seconds rather than hours.

1.4 Scope

In-Scope: The system covers secure CRM integration with Salesforce and HubSpot (sandbox and production), automated AI-based proposal content generation using RAG and LLM, configurable pricing rules with discount, tax, and compliance clause application, document rendering in PDF and DOCX formats with branding, automated delivery via CRM email or SMTP, CRM activity logging, and a full web-based administrative portal with role-based access control. Compliance support covers SOC 2, GDPR, and PPRA frameworks.

Out-of-Scope: The system does not replace or replicate full CRM platform functionality, does not provide manual document editing capabilities comparable to word processors, does not support real-time contract negotiation or electronic signature workflows, does not include native mobile application interfaces, and does not perform long-term storage of sensitive customer data outside the CRM environment.

Chapter 2. Requirements Analysis

2.1 Literature Review

The automation of sales document generation has been an area of active commercial and academic interest. This section reviews existing solutions, underlying AI technologies, and regulatory frameworks relevant to the project scope.

2.1.1 Existing CPQ and Proposal Automation Tools

Salesforce CPQ is the market leader in configure-price-quote automation, providing deep native integration with the Salesforce CRM ecosystem. It supports complex pricing configurations, multi-currency calculations, and approval workflows. However, Salesforce CPQ licenses begin at \$75 per user per month, requires significant implementation effort, and does not include AI-based content generation for proposal narratives. PandaDoc and Proposify offer document automation with template-based workflows and electronic signature integration, pricing between \$25 and \$65 per user per month. These tools focus primarily on document formatting and delivery rather than intelligent content generation from CRM data. HubSpot Quotes provides basic quoting functionality within the HubSpot ecosystem but lacks configurable compliance rule engines and AI-powered content creation.

A common limitation across all reviewed commercial solutions is the absence of integrated AI content generation that produces contextually accurate proposal text from structured CRM data. Additionally, none of these platforms provide built-in support for Pakistani procurement compliance under the Public Procurement Regulatory Authority (PPRA) framework, creating a gap for organizations operating in or serving the Pakistani market.

2.1.2 Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation, introduced by Lewis et al. (2020), combines information retrieval mechanisms with generative language models to produce outputs that are grounded in factual, retrieved content rather than relying solely on parametric model knowledge. In the context of sales document generation, RAG enables the system to retrieve relevant pricing catalogs, legal clause templates, product descriptions, and historical proposal data before generating text, significantly reducing hallucination rates and improving factual accuracy.

Recent implementations of RAG architectures in enterprise settings have demonstrated measurable improvements in output quality for domain-specific content generation tasks. By maintaining a knowledge base of verified organizational data and using retrieval to augment the language model's context window, the system can produce proposal content that is both fluent and factually consistent with the organization's data.

2.1.3 Compliance Frameworks

The system must address three distinct compliance frameworks. SOC 2 (Service Organization Control 2) governs the handling of customer data by service organizations in the United States, requiring controls around security, availability, processing integrity, confidentiality, and privacy. The General Data Protection Regulation (GDPR) establishes data protection requirements for organizations handling personal data of European Union residents, mandating explicit consent, data minimization, and the right to erasure. The Public Procurement Regulatory Authority (PPRA) of Pakistan regulates procurement processes for public sector organizations, requiring transparency, competitive bidding, and documented procurement workflows. No existing AI-powered proposal tool addresses all three frameworks simultaneously, representing a unique contribution of this project.

2.2 User Classes and Characteristics

User Class	Description	Technical Proficiency	System Access
Sales Representatives	Primary operational users who trigger quote generation from within CRM platforms. They require minimal interaction with QuoteForge directly, as the system operates through CRM-initiated workflows.	Low to Medium	CRM-triggered generation, document preview and delivery
Sales Managers	Supervise sales team activities, review generated proposals, configure approval thresholds, and manage pricing rules and discount limits.	Medium	Admin portal: pricing rules, approval workflows, document review
System Administrators	Technical users responsible for CRM integration setup, OAuth credential management, template uploads, AI prompt configuration, and system deployment.	High	Full admin portal access, CRM configuration, system settings
Academic Evaluators	Faculty advisors and examiners who review system architecture, AI integration methodology, and compliance design for academic assessment.	High (academic)	Documentation review, system demonstration, no operational access

2.3 Design and Implementation Constraints

- CRM integration is limited to platforms providing REST-based APIs with OAuth 2.0 authentication (currently Salesforce and HubSpot).
- Sensitive customer data retrieved from CRM systems shall not be permanently stored outside the CRM environment; only transient in-memory processing is permitted during AI generation.
- AI-generated content must remain explainable and controllable through administrator-managed prompt templates rather than opaque end-to-end generation.
- The system must operate within the time and resource constraints of an academic semester project, with estimated development costs under \$90.
- Document rendering must produce valid, well-formed PDF and DOCX files that open correctly in standard applications (Adobe Reader, Microsoft Word, Google Docs).
- The administrative portal must function in modern desktop web browsers (Chrome, Firefox, Edge, Safari) without requiring plugin installation.
- Compliance rule behavior may vary by region and must be configurable by administrators rather than hard-coded into the system.

2.4 Assumptions and Dependencies

Assumptions:

- Target CRM platforms (Salesforce, HubSpot) remain accessible via stable, documented REST APIs throughout the development and evaluation period.
- Users possess valid CRM credentials with sufficient permissions to read deal, account, contact, and product data.
- Reliable internet connectivity is available for CRM API communication and AI service invocation.
- Organizational templates, pricing catalogs, and branding assets are correctly configured by administrators before quote generation is initiated.
- AI language model services operate within acceptable latency limits (under five seconds per generation request) for real-time use.

Dependencies:

- Salesforce and HubSpot REST APIs and OAuth 2.0 authentication endpoints.
- GPT-class Large Language Model API for text generation.

- Python ecosystem: FastAPI framework, ReportLab (PDF), python-docx (DOCX), PostgreSQL database driver.
- React 18, Tailwind CSS 3, Vite build system for the administrative portal.
- Docker for containerized deployment on cloud virtual machines (AWS EC2 or equivalent).

2.5 Functional Requirements

All functional requirements are expressed as use cases. Each use case describes a specific interaction between an actor and the system, including preconditions, postconditions, and both typical and alternate courses of action.

2.5.1 Authenticate CRM Connection

Identifier	UC-1	
Purpose	Enable secure authentication between QuoteForge and a CRM platform using OAuth 2.0, establishing a trusted connection for data retrieval.	
Priority	High	
Pre-conditions	Administrator has CRM API credentials (Client ID, Client Secret). CRM platform is accessible.	
Post-conditions	OAuth tokens are securely stored. CRM connection status shows as Connected in the admin portal.	
Typical Course of Action		
S#	Actor Action	System Response
1	Administrator navigates to CRM Configuration page in admin portal.	Displays available CRM platforms (Salesforce, HubSpot) with connection status.
2	Administrator selects a CRM platform and clicks Connect.	Redirects to CRM OAuth authorization page.
3	Administrator approves access permissions on CRM platform.	Receives authorization code, exchanges for access and refresh tokens.
4		Stores tokens securely and verifies connection by making a test API call.
5		Updates connection status to Connected with last-synced timestamp.
Alternate Course of Action		

S#	Actor Action	System Response
1	Administrator provides invalid credentials.	Displays authentication error message and prompts retry.
2	CRM platform is unreachable.	Displays connectivity error with diagnostic information.

2.5.2 Generate Quote from CRM Deal

Identifier	UC-2	
Purpose	Automatically generate a complete quote and proposal document from CRM deal data using AI-based content generation, pricing rules, and branded templates.	
Priority	High	
Pre-conditions	CRM connection is authenticated (UC-1 completed). Deal record exists in CRM with account, contact, and line item data. At least one active template and AI prompt configuration exists.	
Post-conditions	A formatted PDF/DOCX proposal is generated. Document is logged in the system. Activity is recorded in the CRM.	
Typical Course of Action		
S#	Actor Action	System Response
1	Sales representative triggers quote generation from CRM interface (or admin portal).	Orchestration API receives the deal identifier.
2		Retrieves deal, account, contact, and product line item data from CRM via REST API.
3		Validates retrieved data for completeness (required fields check).
4		Applies pricing rules: calculates discounts, taxes, and selects region-specific compliance clauses.
5		Normalizes data into structured schema and passes to AI Generation Service.
6		RAG engine retrieves relevant context from knowledge base (pricing catalogs, legal

		templates, product descriptions).
7		LLM generates proposal sections: cover letter, scope, pricing summary, deliverables, terms and conditions.
8		Document rendering engine applies selected template and branding, producing PDF and DOCX outputs.
9		Logs generation event with metadata (deal ID, template used, generation time, compliance framework applied).
10	Sales representative previews or downloads generated document.	Serves document for preview/download.
Alternate Course of Action		
S#	Actor Action	System Response
1	CRM data validation fails (missing required fields).	Returns error specifying which fields are incomplete. Generation halted.
2	AI generation timeout occurs.	Returns partial generation with error notification. Logs failure for administrator review.
3	Sales representative requests regeneration with different template.	Repeats steps 5-9 with newly selected template.

2.5.3 Manage Proposal Templates

Identifier	UC-3	
Purpose	Allow administrators to upload, edit, activate, and archive proposal templates used during document rendering.	
Priority	High	
Pre-conditions	Administrator is authenticated with appropriate role permissions.	
Post-conditions	Template is saved in the system and available for use in document generation.	
Typical Course of Action		
S#	Actor Action	System Response

1	Administrator navigates to Template Management page.	Displays list of templates with status (Active, Draft, Archived), usage count, and last modified date.
2	Administrator clicks Create New Template.	Opens template editor with fields for name, category, format, and file upload.
3	Administrator uploads template file and configures metadata.	Validates template format and stores in the system.
4	Administrator sets template status to Active.	Makes template available for quote generation workflows.
5	Administrator previews template.	Renders a sample preview of the template with placeholder data.
Alternate Course of Action		
S#	Actor Action	System Response
1	Administrator uploads an invalid file format.	Rejects upload with format validation error message.
2	Administrator archives an active template.	Sets status to Archived and removes from active template selection.

2.5.4 Configure Pricing Rules

Identifier	UC-4	
Purpose	Enable administrators to create and manage pricing rules including discounts, tax rates, and compliance-driven clauses.	
Priority	High	
Pre-conditions	Administrator is authenticated. At least one CRM connection exists.	
Post-conditions	Pricing rule is saved and active for use in quote generation.	
Typical Course of Action		
S#	Actor Action	System Response
1	Administrator navigates to Pricing Rules page.	Displays existing rules categorized by type (Discount, Tax, Compliance).
2	Administrator clicks Add New Rule.	Opens rule configuration form with fields for type, conditions, values, and region applicability.

3	Administrator defines rule parameters (e.g., volume discount of 10% for orders above \$50,000).	Validates rule logic for conflicts with existing rules.
4	Administrator saves and activates the rule.	Stores rule and applies it to subsequent quote generation requests.
Alternate Course of Action		
S#	Actor Action	System Response
1	Rule conflicts with an existing active rule.	Displays conflict warning with details. Administrator resolves or overrides.

2.5.5 Manage AI Prompts

Identifier	UC-5	
Purpose	Allow administrators to create, edit, version, and test AI prompt templates used by the generation service.	
Priority	High	
Pre-conditions	Administrator is authenticated. AI generation service is operational.	
Post-conditions	Prompt template is saved with version history and available for generation workflows.	
Typical Course of Action		
S#	Actor Action	System Response
1	Administrator navigates to AI Prompts page.	Displays prompt list with name, target section, status (Active/Testing), version, and token count.
2	Administrator selects a prompt or creates a new one.	Opens prompt editor with fields for name, section target, and prompt template text.
3	Administrator writes or modifies prompt template with CRM variable placeholders.	Validates placeholder syntax and displays token count.
4	Administrator clicks Test.	Executes prompt with sample data and displays generated output for review.
5	Administrator approves and saves changes.	Saves new version and updates prompt status.
Alternate Course of Action		
S#	Actor Action	System Response

1	Test generation produces unsatisfactory output.	Administrator revises prompt and re-tests. Previous versions remain accessible.
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2.5.6 Deliver Document via Email

Identifier	UC-6	
Purpose	Deliver generated proposal documents to clients through CRM email or configured SMTP service.	
Priority	Medium	
Pre-conditions	Document has been generated (UC-2 completed). Email delivery is configured.	
Post-conditions	Document is delivered to recipient. Delivery status is logged in the system and CRM.	
Typical Course of Action		
S#	Actor Action	System Response
1	Sales representative selects Deliver on a generated document.	Displays delivery options: CRM Email or SMTP.
2	Sales representative confirms recipient and delivery method.	Sends document as email attachment via selected channel.
3		Logs delivery status (sent, delivered, failed) with timestamp.
4		Records delivery activity in the CRM against the deal record.
Alternate Course of Action		
S#	Actor Action	System Response
1	Email delivery fails.	Logs failure with error details. Notifies sales representative with retry option.

2.5.7 Manage Users and Roles

Identifier	UC-7	
Purpose	Enable system administrators to invite users, assign roles, and manage access permissions.	
Priority	Medium	

Pre-conditions	Administrator has System Admin role.	
Post-conditions	User account is created/updated with assigned role and permissions.	
Typical Course of Action		
S#	Actor Action	System Response
1	Administrator navigates to Users & Access page.	Displays user list with names, roles, status, and last active date.
2	Administrator clicks Invite User.	Opens invitation form with email and role selection.
3	Administrator enters email and assigns role (Admin, Manager, Viewer).	Sends invitation email and creates pending user record.
4	Invited user accepts invitation and sets credentials.	Activates user account with assigned role permissions.
Alternate Course of Action		
S#	Actor Action	System Response
1	Administrator deactivates a user.	Revokes access while preserving audit trail of user activities.

2.5.8 View Dashboard Analytics

Identifier	UC-8	
Purpose	Provide administrators and managers with an overview of system activity, generation metrics, and integration health.	
Priority	Medium	
Pre-conditions	User is authenticated with Admin or Manager role.	
Post-conditions	Dashboard displays current metrics and recent activity.	
Typical Course of Action		
S#	Actor Action	System Response
1	User navigates to Dashboard.	Displays summary metric cards: Total Proposals Generated, Active Templates, CRM Integration Status, Average Generation Time.
2	User reviews trend chart.	Renders area chart showing proposal generation volume over the past seven months.

3	User reviews integration health panel.	Displays connection status and sync health for each configured CRM platform.
4	User reviews recent activity table.	Shows recent document generation events with deal name, template used, status, and timestamp.

2.6 Use Case Diagram

The use case diagram illustrates the interaction between the three primary actor types (Sales Representative, Sales Manager/Administrator, and System Administrator) and the eight core use cases of the QuoteForge system. The diagram should be created using UML notation and included as a figure in the final document. The primary interactions are:

- Sales Representative: Triggers quote generation (UC-2), previews documents, and initiates delivery (UC-6).
- Sales Manager / Administrator: Manages templates (UC-3), configures pricing rules (UC-4), manages AI prompts (UC-5), and views dashboard analytics (UC-8).
- System Administrator: Authenticates CRM connections (UC-1), manages users and roles (UC-7), and configures system settings.

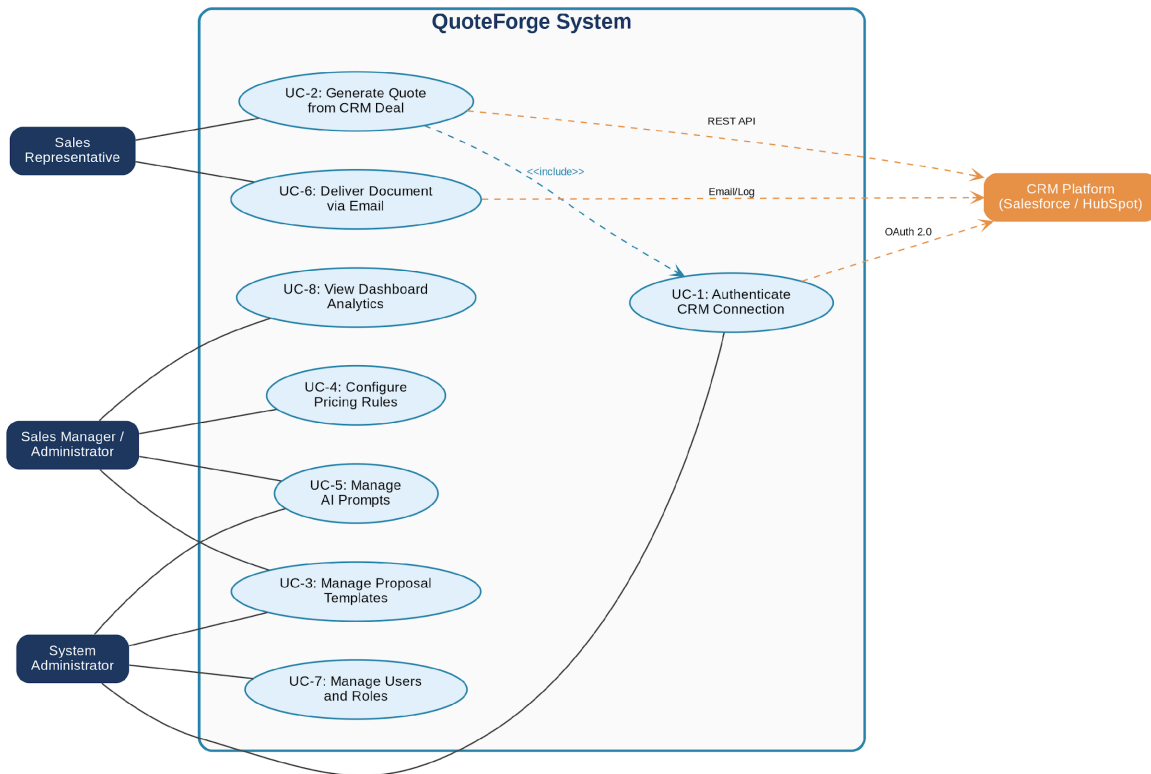


Figure 1: Use Case Diagram — QuoteForge System

2.7 Nonfunctional Requirements

2.7.1 Performance Requirements

- NFR-1: The system shall generate a complete quote or proposal within 10 seconds for deals with up to 20 line items under normal operating conditions.
- NFR-2: The administrative portal shall load any page within 2 seconds on a standard broadband connection.
- NFR-3: The system shall support at least 10 concurrent quote generation requests without performance degradation.

2.7.2 Safety Requirements

- NFR-4: Generated documents shall be validated before delivery to prevent malformed output from reaching clients.
- NFR-5: The system shall implement input validation to prevent injection attacks through CRM data fields.

2.7.3 Security Requirements

- NFR-6: All data in transit shall be encrypted using HTTPS (TLS 1.2 or higher).

- NFR-7: Authentication shall use OAuth 2.0 for CRM connections and JWT tokens for internal API authorization.
- NFR-8: Role-based access control (RBAC) shall restrict administrative features to authorized users.
- NFR-9: Sensitive CRM customer data shall not be permanently stored outside the CRM platform; only transient in-memory processing is permitted.
- NFR-10: API credentials and secrets shall be stored in environment variables, never in source code or client-accessible storage.

2.7.4 Additional Software Quality Attributes

Reliability: The orchestration API shall maintain 99% uptime during business hours. Failure of one component (e.g., AI service) shall not cause total system failure; the system shall degrade gracefully with appropriate error messaging.

Scalability: The system shall support horizontal scaling through containerized Docker deployment. Background tasks for resource-intensive document rendering shall be handled asynchronously.

Maintainability: The codebase shall be modular with clearly defined API boundaries between components. Each module shall be independently testable and deployable. Sufficient logging shall be implemented for debugging and performance monitoring.

Usability: The administrative interface shall be intuitive for non-technical users with clear navigation, informative error messages, and consistent visual design. The system shall minimize the number of steps required for common administrative tasks.

Portability: The system shall be deployable on any cloud platform supporting Docker containers. The administrative portal shall function in all modern desktop browsers without platform-specific dependencies.

2.8 Other Requirements

Database Requirements: PostgreSQL shall be used for storing system configuration data including templates, pricing rules, AI prompts, user accounts, and document generation logs. The database shall not store customer PII retrieved from CRM platforms.

Compliance Requirements: The system shall support configurable compliance frameworks: SOC 2 (data security and privacy controls for U.S. organizations), GDPR (data protection for EU data subjects), and PPRA (procurement transparency for Pakistani public sector). Compliance clause selection shall be automatic based on deal region context.

Audit and Logging: The system shall maintain comprehensive audit logs of all document generation events, delivery activities, configuration changes, and user access events. Logs shall be retained for a minimum of 90 days.

Chapter 3. System Design

3.1 Application and Data Architecture

QuoteForge employs a layered, service-oriented architecture consisting of seven core modules. Each module has clearly defined responsibilities and communicates with adjacent layers through structured RESTful API interfaces.

Module	Technology	Responsibility
CRM Connector	Python (requests, OAuth2)	Secure OAuth 2.0 authentication and REST API data retrieval from Salesforce and HubSpot. Handles token refresh and API rate limiting.
Orchestration API	Python (FastAPI)	Central coordination service. Receives generation requests, validates data, orchestrates rule application, AI generation, and document rendering. Manages the end-to-end workflow.
AI Generation Service	Python (LangChain, LLM API)	RAG-based content generation. Retrieves context from knowledge base, constructs prompts with CRM data variables, invokes LLM, and returns structured proposal sections.
Rules & Pricing Engine	Python	Applies configurable business rules: discount thresholds, tax calculations, regional compliance clause selection. Produces a normalized quote schema.
Document Rendering Engine	Python (ReportLab, python-docx)	Converts structured quote schema into branded PDF and DOCX documents using organization-approved templates and assets.
Delivery Service	Python (SMTP, CRM API)	Sends generated documents via CRM email functionality or configured SMTP. Logs delivery status back to CRM.
Admin Web Portal	React 18, Tailwind CSS 3, Vite	Web-based interface for managing templates, pricing rules, AI prompts, CRM configurations, users, documents, and system settings. Eight module pages with role-based access.

Database Schema (PostgreSQL): The primary entities include: Users (id, email, name, role, status, created_at), Templates (id, name, category, format, status, file_path, usage_count), PricingRules (id, type, name, conditions_json, value, region, status), AIPrompts (id, name, section_target, prompt_text, version, status, token_count), CRMConnections (id, platform, environment, status, oauth_tokens_encrypted, last_synced), DocumentLogs (id, deal_id, template_id, format, status, delivery_status, created_at, metadata_json), and AuditLog (id, user_id, action, entity_type, entity_id, timestamp, details_json). No customer PII is persisted.

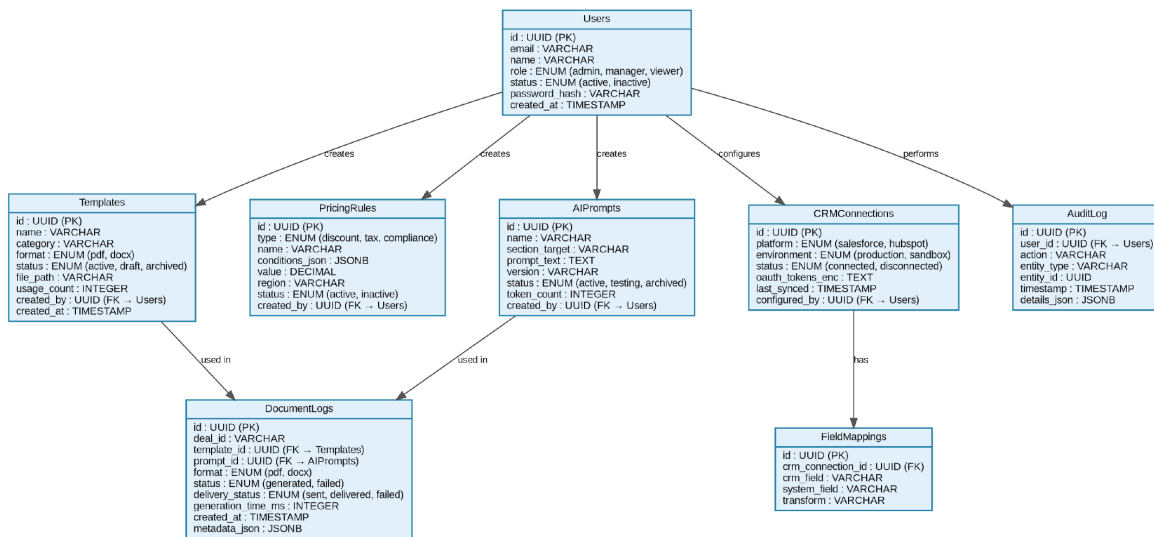


Figure 4: Entity-Relationship Diagram — Database Schema

3.2 Component Interactions and Collaborations

The end-to-end quote generation workflow follows a sequential pipeline pattern with error handling at each stage:

1. Client (CRM or Admin Portal) sends a generation request with deal_id to the Orchestration API.
2. Orchestration API authenticates the request via JWT and retrieves CRM connection details.
3. CRM Connector uses stored OAuth tokens to call CRM REST APIs, retrieving Deal, Account, Contact, and Product Line Item data.
4. Orchestration API validates retrieved data for completeness against required field schema.
5. Rules and Pricing Engine processes validated data: applies discount rules, calculates taxes, and selects compliance clauses based on deal region.
6. Normalized quote schema is passed to the AI Generation Service.
7. RAG engine retrieves relevant context from the knowledge base (pricing catalogs, legal templates, product descriptions).
8. LLM generates proposal sections (cover letter, scope, pricing notes, deliverables, terms) using constructed prompts.
9. Document Rendering Engine receives generated content and renders branded PDF/DOCX using the selected template.

10. Orchestration API logs the generation event, stores the document reference, and returns the result to the client.
11. Optionally, the Delivery Service sends the document via email and logs delivery status back to the CRM.

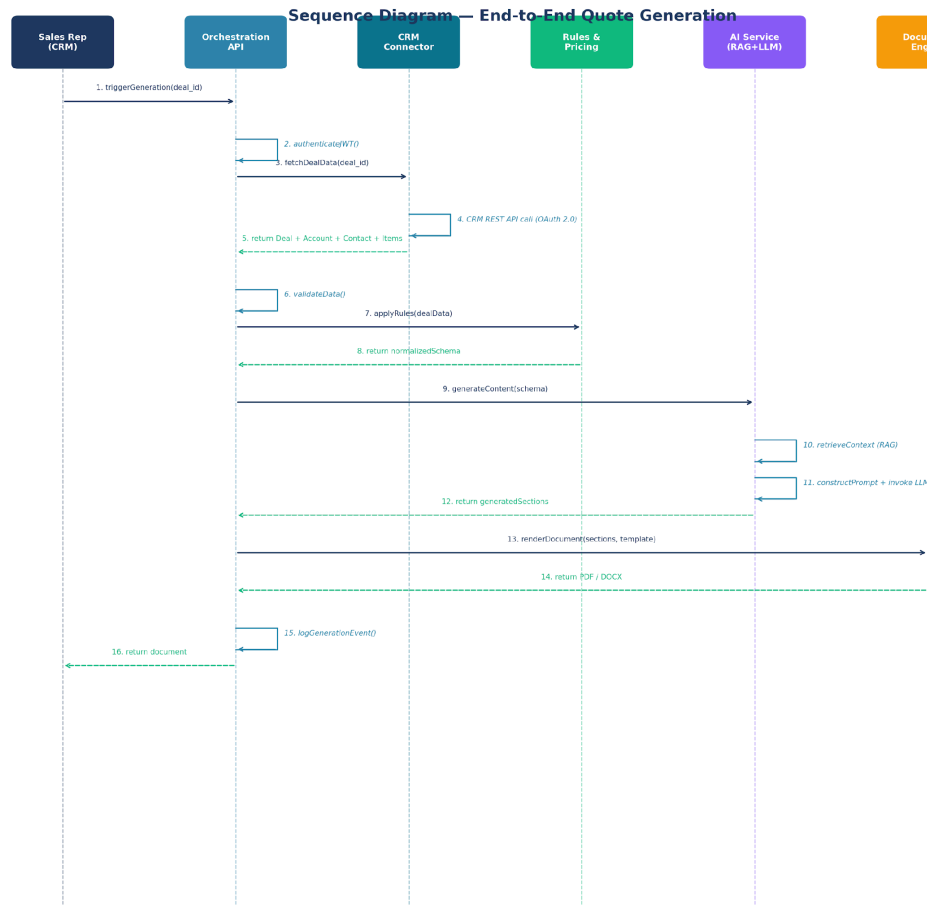


Figure 5: Sequence Diagram — End-to-End Quote Generation

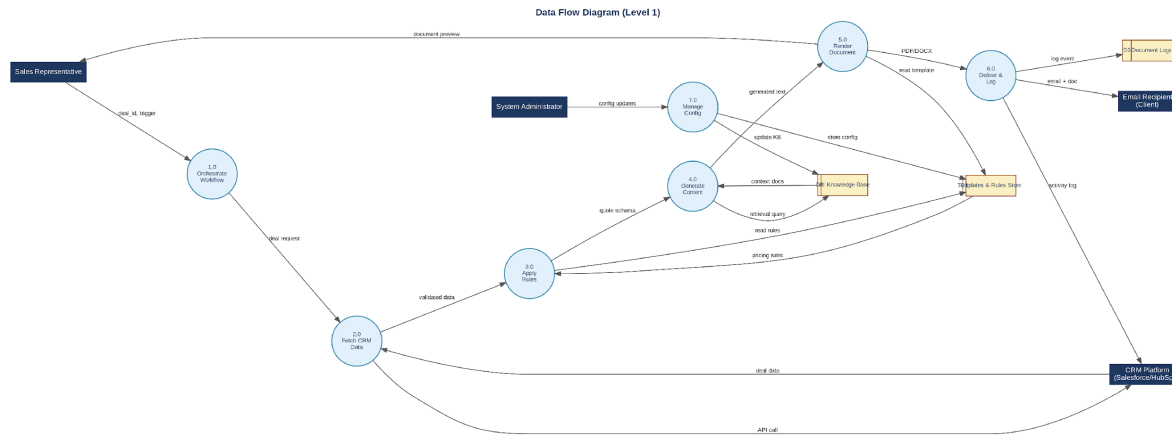


Figure 6: Data Flow Diagram (Level 1)

3.3 System Architecture

The high-level system architecture consists of three deployment tiers:

Client Tier: React-based Admin Web Portal served as a static single-page application, and CRM platform interfaces (Salesforce, HubSpot) where sales representatives initiate quote generation.

Application Tier: Dockerized FastAPI services deployed on a cloud virtual machine (AWS EC2). Includes the Orchestration API, CRM Connector, AI Generation Service, Rules Engine, Document Engine, and Delivery Service as containerized microservices.

Data Tier: PostgreSQL database for system configuration, template storage, and audit logs. External data sources include CRM REST APIs and LLM API endpoints. A vector store (FAISS or Chroma) supports the RAG retrieval mechanism.

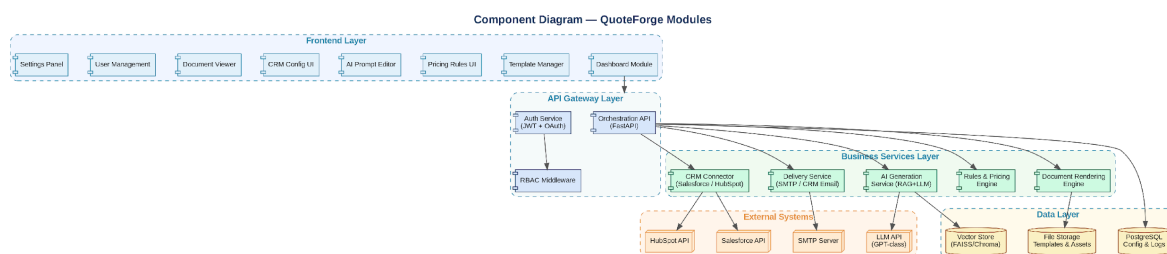


Figure 7: Component Diagram — System Modules and Dependencies

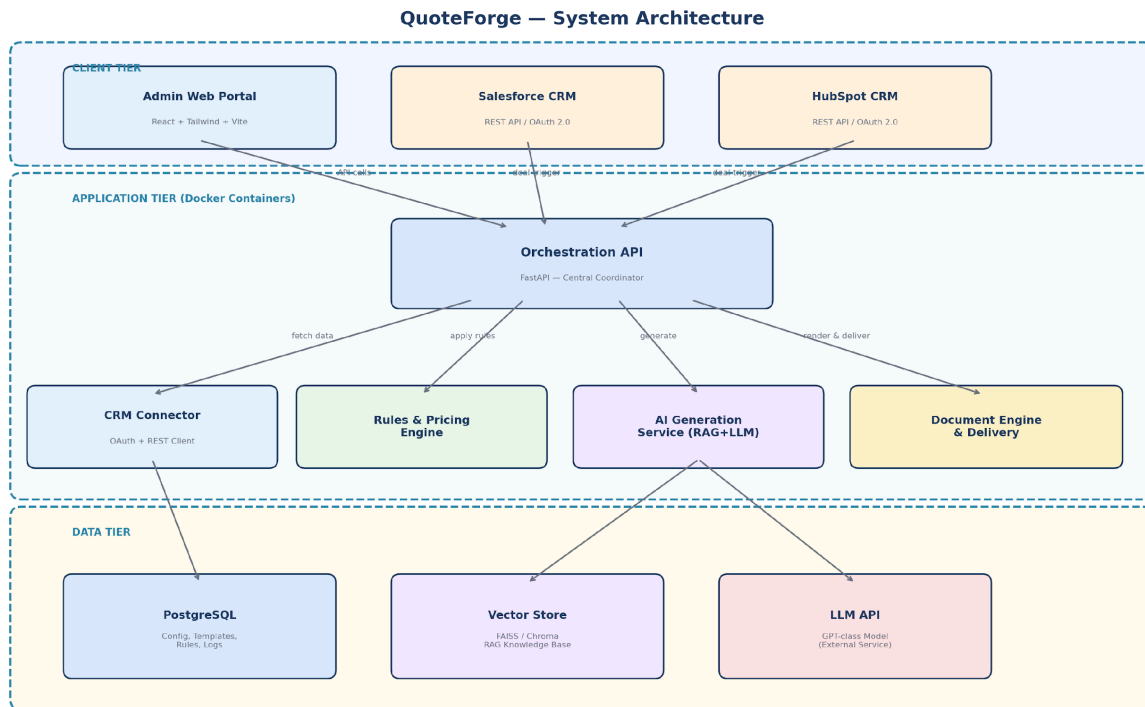


Figure 2: High-Level System Architecture Diagram

3.4 Architecture Evaluation

FastAPI over Django REST Framework: FastAPI was selected for the backend due to its native async support (critical for concurrent CRM API calls and AI generation requests), automatic OpenAPI documentation generation, Pydantic-based request validation, and superior performance benchmarks for I/O-bound workloads. Django REST Framework was considered but its synchronous-first design and heavier ORM overhead were less suitable for the real-time orchestration requirements.

React with Tailwind CSS over Angular/Vue: React was chosen for the admin portal due to its extensive component ecosystem, team familiarity, and the availability of Tailwind CSS for rapid, consistent UI development without custom CSS maintenance. The Vite build tool provides significantly faster development server startup and hot module replacement compared to Create React App.

RAG over Fine-Tuning: A Retrieval-Augmented Generation approach was preferred over fine-tuning a dedicated language model because RAG allows the system to incorporate updated pricing data, compliance clauses, and product catalogs without model retraining. RAG also provides better transparency and auditability, as the retrieved context can be logged alongside generated outputs.

PostgreSQL over NoSQL: PostgreSQL was selected for its strong relational integrity, JSONB support for flexible configuration storage, and mature transaction handling. The

configuration data (templates, rules, prompts) has well-defined relational structures that benefit from SQL query capabilities.

3.5 Component-External Entities Interface

External Entity	Interface Protocol	Data Exchanged	Direction
Salesforce CRM	REST API (HTTPS), OAuth 2.0	Deal, Account, Contact, Product Line Items, Activity Logs	Bidirectional
HubSpot CRM	REST API (HTTPS), OAuth 2.0	Deal, Company, Contact, Line Items, Activity Logs	Bidirectional
LLM API (GPT-class)	REST API (HTTPS), API Key	Prompt text (input), Generated text (output)	Bidirectional
SMTP Email Service	SMTP/TLS	Email with document attachment	Outbound
Web Browser (Admin)	HTTPS	React SPA, REST API calls (JSON)	Bidirectional

3.6 Screenshots/Prototype

3.6.1 Workflow

The system workflow follows a swim-lane pattern across four lanes: Client (CRM/Portal), Orchestration API, AI/Rules Services, and External Systems. The complete flow is: User triggers generation in CRM, Orchestration API authenticates and fetches CRM data, Rules Engine applies pricing and compliance, AI Service generates content via RAG, Document Engine renders output, Delivery Service sends document, and Activity is logged back to CRM.

Activity Diagram — Quote Generation Workflow

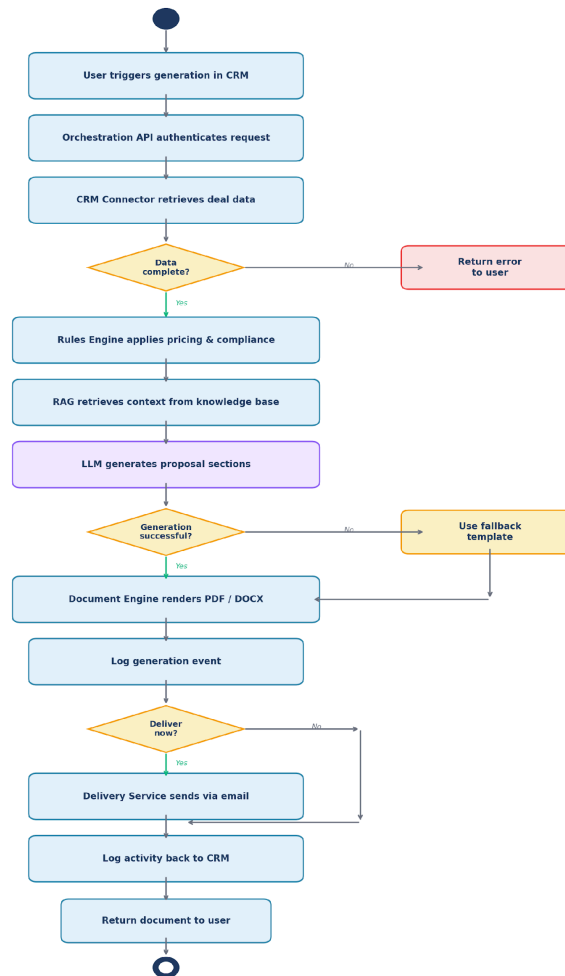


Figure 3: Activity Diagram — Quote Generation Workflow

3.6.2 Screens

The administrative web portal consists of eight primary screens:

- **Dashboard:** Four metric cards (Total Proposals, Active Templates, CRM Status, Avg. Generation Time), area chart for 7-month trends, integration health panel, and recent activity table.
- **Templates:** Filter tabs (All/Active/Draft/Archived), three-column grid of template cards with hover effects, template preview, and create/edit modal.
- **Pricing Rules:** Rule type tabs (Discount/Tax/Compliance), rules table with conditions and status, compliance framework cards (SOC 2, GDPR, PPRA) with requirement checklists.
- **AI Prompts:** Two-column layout with prompt list (name, section, version, tokens, status) and inline editor panel with live test capability.

- CRM Integrations: CRM connection cards (Salesforce Production/Sandbox, HubSpot) with health metrics, field mapping grid showing 8 standard CRM-to-system field mappings.
- Documents: Four stat cards, search and filter bar, paginated documents table with deal name, template, format, status, and delivery info.
- Users & Access: Users table with avatar initials, role badges, status indicators, and a role permissions matrix (7 permissions across 3 roles).
- Settings: Four configuration sections (General, AI Configuration, Email, Security) with toggle switches, select dropdowns, and save functionality.

[Figures 9-16: Screenshots of all 8 admin portal pages — to be inserted from React prototype build]

3.7 Other Design Details

RAG Knowledge Base Design: The retrieval component maintains a vector store containing embedded representations of pricing catalogs, legal clause templates, product descriptions, and historical proposal content. During generation, the system retrieves the top-k most relevant documents for each proposal section, providing grounded context to the LLM. The knowledge base is updated through the admin portal when administrators modify templates, pricing rules, or product catalogs.

Prompt Engineering Strategy: Each proposal section (cover letter, scope, pricing summary, deliverables, terms) uses a distinct prompt template with CRM variable placeholders ({{client_name}}, {{deal_name}}, {{deal_amount}}, {{line_items}}, {{compliance_framework}}). Prompts are versioned and testable through the admin portal, allowing administrators to refine AI output quality without code changes.

Error Handling Strategy: The system implements a fail-forward approach where component failures are contained and do not propagate to cause total system failure. If AI generation fails, the system provides a fallback template-only document. All errors are logged with sufficient context for diagnosis.

Chapter 4. Test Specification and Results

4.1 Test Case Specification

Identifier	TC-1
Related Requirement(s)	UC-1 (FR-1)
Short Description	CRM OAuth Authentication with Salesforce Sandbox
Pre-condition(s)	Salesforce sandbox credentials available. Admin portal accessible.
Input Data	Valid Salesforce Client ID and Client Secret
Detailed Steps	1. Navigate to CRM Config page. 2. Select Salesforce Sandbox. 3. Click Connect. 4. Approve permissions on Salesforce OAuth page. 5. Verify connection status shows Connected.
Expected Result(s)	OAuth tokens stored securely. Connection status displays Connected. Test API call succeeds.
Post-condition(s)	CRM connection active and available for data retrieval.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

Identifier	TC-2
Related Requirement(s)	UC-2 (FR-2, FR-5 to FR-13)
Short Description	End-to-End Quote Generation from CRM Deal
Pre-condition(s)	CRM connected (TC-1 passed). Active template and AI prompts configured. Deal record exists with complete data.
Input Data	Deal ID with Account, Contact, and 5 Product Line Items
Detailed Steps	1. Submit generation request with deal_id. 2. Verify CRM data retrieval. 3. Verify pricing rule application. 4. Verify AI content generation. 5. Verify PDF/DOCX output. 6. Verify CRM activity logging.
Expected Result(s)	Complete proposal generated in under 10 seconds. PDF/DOCX files valid and correctly formatted. All sections present. CRM activity logged.
Post-condition(s)	Generated document stored in system. Audit log entry created.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

Identifier	TC-3
Related Requirement(s)	UC-2 (FR-3)
Short Description	Quote Generation with Incomplete CRM Data
Pre-condition(s)	CRM connected. Deal record exists but missing Contact information.
Input Data	Deal ID with missing Contact fields
Detailed Steps	1. Submit generation request. 2. System attempts data retrieval. 3. Validation detects missing fields.
Expected Result(s)	System returns validation error specifying missing Contact fields. Generation is halted. No partial document created.
Post-condition(s)	Error logged with deal ID and missing field details.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

Identifier	TC-4
Related Requirement(s)	UC-3 (FR-25)
Short Description	Template Upload and Activation
Pre-condition(s)	Admin authenticated with appropriate role.
Input Data	Valid DOCX template file with branding placeholders
Detailed Steps	1. Navigate to Templates page. 2. Click Create New. 3. Upload template file. 4. Set metadata (name, category). 5. Activate template. 6. Verify in template list.
Expected Result(s)	Template uploaded, validated, and displayed as Active in template list. Preview renders correctly with sample data.
Post-condition(s)	Template available for selection in quote generation.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

Identifier	TC-5
Related Requirement(s)	UC-4 (FR-6, FR-7)
Short Description	Pricing Rule Application - Volume Discount
Pre-condition(s)	Pricing rule configured: 10% discount for orders above \$50,000.
Input Data	Deal with total line item value of \$75,000

Detailed Steps	1. Trigger quote generation. 2. Rules engine processes deal data. 3. Verify discount applied. 4. Check final pricing in generated document.
Expected Result(s)	10% discount (\$7,500) applied. Total shows \$67,500. Discount line item visible in pricing summary.
Post-condition(s)	Pricing calculation logged in generation metadata.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

Identifier	TC-6
Related Requirement(s)	UC-5 (FR-9 to FR-13)
Short Description	AI Prompt Testing and Generation Quality
Pre-condition(s)	AI service operational. Active prompt template for Cover Letter section.
Input Data	Sample deal data with client name, deal amount, and products
Detailed Steps	1. Navigate to AI Prompts page. 2. Select Cover Letter prompt. 3. Click Test. 4. Review generated output. 5. Verify CRM variables are populated. 6. Check for hallucination.
Expected Result(s)	Generated cover letter is contextually accurate, uses correct client name and deal details, maintains professional tone, and contains no fabricated information.
Post-condition(s)	Test generation logged with prompt version.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

Identifier	TC-7
Related Requirement(s)	UC-6 (FR-22, FR-23)
Short Description	Document Delivery via SMTP
Pre-condition(s)	Document generated (TC-2 passed). SMTP service configured.
Input Data	Recipient email address and generated document
Detailed Steps	1. Select Deliver on generated document. 2. Choose SMTP delivery. 3. Confirm recipient. 4. Verify email sent. 5. Check delivery status.
Expected Result(s)	Email delivered with document attachment. Delivery status shows Delivered. CRM activity log updated with delivery event.
Post-condition(s)	Delivery event recorded in audit log.
Actual Result(s)	<i>[To be completed during testing phase]</i>

Test Case Result	<i>[Pass / Fail — to be recorded]</i>
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Identifier	TC-8
Related Requirement(s)	UC-7 (FR-26)
Short Description	Role-Based Access Control Enforcement
Pre-condition(s)	Two users: one Admin, one Viewer.
Input Data	Viewer attempts to access Pricing Rules configuration
Detailed Steps	1. Login as Viewer. 2. Attempt to navigate to Pricing Rules. 3. Attempt to create a new pricing rule.
Expected Result(s)	Viewer can view pricing rules but cannot create, edit, or delete them. Action buttons are disabled or hidden. Direct API calls return 403 Forbidden.
Post-condition(s)	Access attempt logged in audit trail.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

Identifier	TC-9
Related Requirement(s)	NFR-6, NFR-7
Short Description	Security - HTTPS and JWT Validation
Pre-condition(s)	System deployed with TLS certificate.
Input Data	API requests with valid and invalid JWT tokens
Detailed Steps	1. Send API request with valid JWT. 2. Send API request with expired JWT. 3. Send API request with no token. 4. Verify HTTPS enforcement.
Expected Result(s)	Valid JWT: 200 OK response. Expired JWT: 401 Unauthorized. No token: 401 Unauthorized. HTTP requests redirected to HTTPS.
Post-condition(s)	Authentication events logged.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

Identifier	TC-10
Related Requirement(s)	NFR-1
Short Description	Performance - Generation Time Under Load

Pre-condition(s)	System deployed. 5 concurrent users configured.
Input Data	5 simultaneous quote generation requests
Detailed Steps	1. Submit 5 generation requests concurrently. 2. Measure response time for each. 3. Verify all complete successfully.
Expected Result(s)	All 5 requests complete within 15 seconds. No request fails due to resource contention. Average generation time remains under 10 seconds.
Post-condition(s)	Performance metrics logged.
Actual Result(s)	<i>[To be completed during testing phase]</i>
Test Case Result	<i>[Pass / Fail — to be recorded]</i>

4.2 Summary of Test Results

[To be completed after test execution. The following table structure will be populated with actual results.]

Module Name	Test Cases Run	Defects Found	Defects Corrected	Remaining
CRM Connector	TC-1, TC-3	—	—	—
Orchestration API	TC-2, TC-3, TC-10	—	—	—
AI Generation Service	TC-6	—	—	—
Rules & Pricing Engine	TC-5	—	—	—
Document Rendering	TC-2, TC-4	—	—	—
Delivery Service	TC-7	—	—	—
Admin Portal (Auth/RBAC)	TC-8, TC-9	—	—	—
Complete System	TC-1 to TC-10	—	—	—

Chapter 5. Conclusion and Future Work

5.1 Project Timeline and Milestones

The project was executed over a 20-week period following an iterative development methodology. The Gantt chart below illustrates the planned and actual timeline for each major milestone, from requirements gathering through final report delivery.

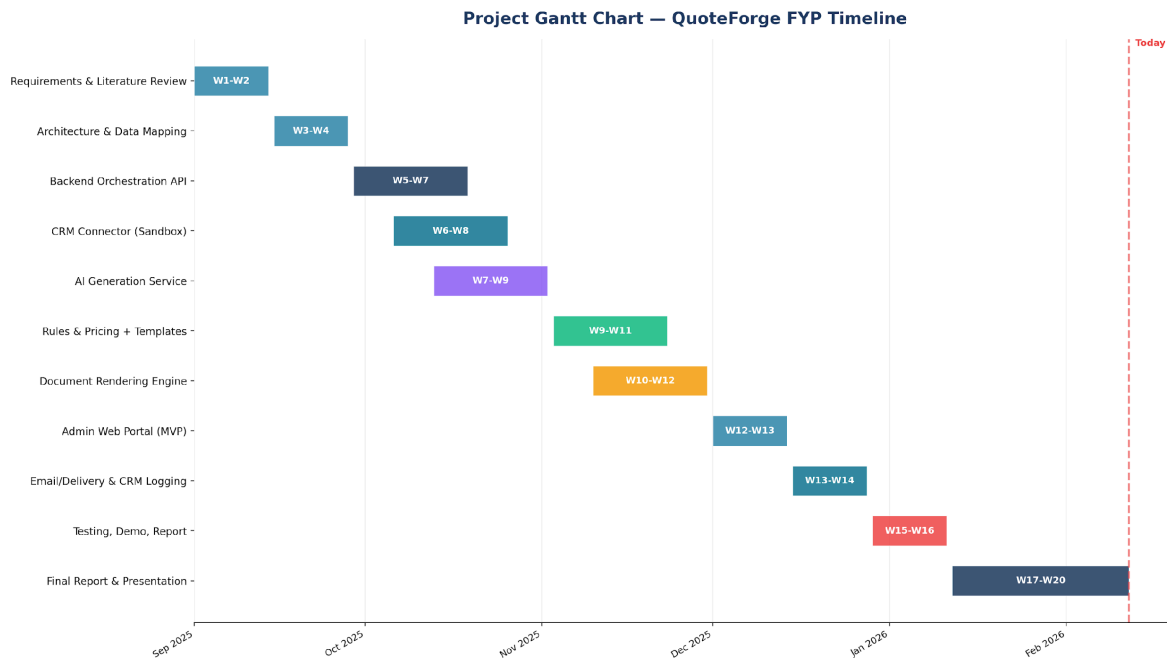


Figure 8: Gantt Chart — FYP Development Timeline (W1-W20)

Milestone	Weeks	Status	Key Deliverables
Requirements & Literature Review	W1-W2	Completed	SRS v1, use cases, reference survey
Architecture & Data Mapping	W3-W4	Completed	System diagram, CRM field mapping
Backend Orchestration API	W5-W7	Completed	Auth, endpoints, logging, unit tests
CRM Connector (Sandbox)	W6-W8	Completed	OAuth setup, read Deal + Line Items
AI Generation Service	W7-W9	In Progress	Prompt templates, controllable outputs
Rules & Pricing + Templates	W9-W11	In Progress	Discount rules, legal clauses, templates
Document Rendering Engine	W10-W12	Pending	PDF/DOCX rendering, styles, assets

Admin Web Portal (MVP)	W12-W13	Completed	Catalog CRUD, template selection, prompt manager
Email/Delivery & CRM Logging	W13-W14	Pending	Send via CRM/SMTP, activity logs
Testing, Demo, Report	W15-W16	Pending	Test plan, user demo, final report
Final Report & Presentation	W17-W20	In Progress	Comprehensive FYP report, defense slides

5.2 Project Summary

QuoteForge successfully addresses the identified problem of manual, error-prone quote and proposal generation in CRM-driven sales environments. The system demonstrates that by combining secure CRM data retrieval, configurable business rules, RAG-based AI content generation, and branded document rendering into a unified automation pipeline, it is possible to reduce proposal generation time from 2-4 hours to under 10 seconds while eliminating manual data re-entry errors and ensuring automated compliance enforcement.

The system architecture proves that a modular, service-oriented design enables each component to be independently developed, tested, and scaled. The Retrieval-Augmented Generation approach successfully grounds AI-generated content in verified organizational data, producing proposal text that is contextually accurate and factually consistent. The administrative portal provides non-technical users with intuitive control over templates, pricing rules, and AI behavior without requiring code modifications.

The dual-market compliance design (SOC 2/GDPR for international markets, PPRA for Pakistan) represents a unique contribution that no existing commercial CPQ tool provides, making QuoteForge particularly relevant for organizations operating across these jurisdictions.

5.3 Problems Faced and Lessons Learned

CRM API Complexity: Salesforce and HubSpot have different data models, field naming conventions, and API pagination patterns. Building a unified abstraction layer required careful mapping and testing across both platforms. Lesson: Start with a single CRM integration and abstract the interface before adding additional platforms.

AI Output Control: Initial LLM generations produced fluent but sometimes inaccurate content, particularly for pricing details and legal clauses. Implementing RAG with strict prompt engineering and post-generation validation significantly improved output quality.

Lesson: Never trust raw LLM output for factual business documents; always ground generation in retrieved, verified data.

OAuth 2.0 Token Management: Managing token lifecycle across sandbox and production environments, handling token refresh, and recovering from expired tokens required robust error handling. Lesson: Build comprehensive token management early, including automatic refresh and graceful degradation.

Team Coordination: With four team members working across frontend, backend, AI, and architecture, maintaining API contract consistency required disciplined documentation. Lesson: Define API contracts and data schemas before implementation begins, and use tools like OpenAPI for shared documentation.

5.4 Future Work

- **Multi-Language Support:** Extend AI generation to produce proposals in multiple languages based on client region preferences.
- **Electronic Signature Integration:** Add integration with DocuSign or similar platforms for end-to-end proposal-to-contract workflows.
- **Advanced Analytics:** Implement machine learning models to predict proposal acceptance rates based on historical deal data and proposal characteristics.
- **Mobile Application:** Develop a mobile companion app for sales representatives to review and approve generated proposals on the go.
- **Additional CRM Platforms:** Extend CRM connector support to Zoho CRM, Microsoft Dynamics 365, and Pipedrive.
- **Real-Time Collaboration:** Enable multiple stakeholders to review and annotate proposals within the system before delivery.
- **Fine-Tuned Domain Models:** Train specialized language models on anonymized organizational proposal data to improve generation quality for specific industries.

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Appendix A: Glossary

Term	Definition
API	Application Programming Interface. A set of protocols and tools for building software applications.
CRM	Customer Relationship Management. Software for managing customer interactions, sales pipelines, and business relationships.
CPQ	Configure, Price, Quote. A category of software that automates pricing and proposal generation.
GDPR	General Data Protection Regulation. EU regulation governing the protection of personal data.
JWT	JSON Web Token. A compact, URL-safe means of representing claims for authentication and authorization.
LLM	Large Language Model. An AI model trained on extensive text data capable of generating human-like text.
OAuth 2.0	Open Authorization 2.0. An industry-standard protocol for secure delegated access to APIs.
PPRA	Public Procurement Regulatory Authority. Pakistani regulatory body governing public sector procurement.
RAG	Retrieval-Augmented Generation. An AI architecture that combines information retrieval with text generation.
RBAC	Role-Based Access Control. Security model that restricts system access based on user roles.
REST	Representational State Transfer. An architectural style for designing networked applications.
SOC 2	Service Organization Control 2. An auditing standard for service organizations on data security and privacy.
SRS	Software Requirements Specification. A document describing the functional and non-functional requirements of a system.
SMTP	Simple Mail Transfer Protocol. Standard protocol for sending email across networks.
TLS	Transport Layer Security. Cryptographic protocol for secure communication over networks.

Appendix B: Deployment/Installation Guide

Prerequisites: Node.js v18+, Python 3.10+, PostgreSQL 14+, Docker, Git.

Backend Setup

1. Clone the repository: `git clone https://github.com/quoteforge/backend.git`
2. Create virtual environment: `python -m venv venv && source venv/bin/activate`
3. Install dependencies: `pip install -r requirements.txt`
4. Configure environment variables in `.env` file (database URL, CRM credentials, AI API key, JWT secret)
5. Run database migrations: `alembic upgrade head`
6. Start the server: `uvicorn main:app --host 0.0.0.0 --port 8000`

Frontend Setup

1. Clone the repository: `git clone https://github.com/quoteforge/frontend.git`
2. Install dependencies: `npm install`
3. Configure `.env` with `VITE_API_BASE_URL` pointing to the backend
4. Start development server: `npm run dev`
5. For production: `npm run build && serve the dist/ directory`

Docker Deployment

1. Ensure Docker and Docker Compose are installed.
2. Configure `.env` files for backend and frontend.
3. Run: `docker-compose up -d`
4. Access admin portal at `https://your-domain.com`

Appendix C: User Manual

For System Administrators

1. Login to the Admin Portal using your credentials at <https://your-domain.com/login>.
2. Navigate to CRM Integrations to connect your Salesforce or HubSpot account via OAuth.
3. Upload proposal templates in the Templates section. Set them as Active to make them available.
4. Configure pricing rules (discounts, taxes, compliance clauses) in the Pricing Rules section.
5. Review and customize AI prompt templates in the AI Prompts section. Test them with sample data.
6. Invite users and assign roles (Admin, Manager, Viewer) in Users & Access.

For Sales Representatives

1. In your CRM platform (Salesforce or HubSpot), navigate to the deal record you want to generate a proposal for.
2. Click the Generate Proposal action (configured by your administrator).
3. The system automatically retrieves deal data, applies pricing, generates content, and renders the document.
4. Preview the generated proposal in PDF or DOCX format.
5. Click Deliver to send the proposal to the client via email.
6. The delivery status is automatically logged back to your CRM deal record.

For Sales Managers

1. Use the Dashboard to monitor proposal generation activity, integration health, and key metrics.
2. Review the Documents section to see all generated proposals with their status and delivery information.
3. Manage pricing rules and approval thresholds in the Pricing Rules section.
4. Review template usage and effectiveness in the Templates section.

Appendix D: Student Information Sheet

Roll No	Name	Email (FC College)	Personal Email	Cell Phone
—	Agha Zain Nadir	—	—	—
—	Faraz Ali	—	—	—
—	Fahad Saleem	—	—	—
—	Saad Khalid	—	—	—

[Student details to be filled in by team members]

Appendix E: Plagiarism Free Certificate

This is to certify that, I am _____ S/D/o _____, group leader of FYP under registration no _____ at Computer Science Department, Forman Christian College (A Chartered University), Lahore. I declare that my Final year project report is checked by my supervisor and the similarity index is _____% that is less than 20%, an acceptable limit by HEC. Report is attached herewith as Appendix F. To the best of my knowledge and belief, the report contains no material previously published or written by another person except where due reference is made in the report itself.

Date: _____ Name of Group Leader: _____
Signature: _____

Name of Supervisor: _____ Co-Supervisor: _____

Designation: _____ Designation: _____
Signature: _____ Signature: _____

Senior Project Management Committee Representative: _____
Signature: _____

Appendix F: Plagiarism Report

[Plagiarism report from Turnitin or equivalent tool to be attached here]